

```

> restart
> edo := y'' + 0.447^2 * y = 0;
      edo :=  $\frac{d^2}{dx^2} y(x) + 0.199809 y(x) = 0$  (1)
> cond := y(0) = 1, D(y)(0) = 2;
      cond := y(0) = 1, D(y)(0) = 2 (2)
> sol := dsolve({edo, cond}, y(x));
      sol :=  $y(x) = \frac{2000}{447} \sin\left(\frac{447}{1000} x\right) + \cos\left(\frac{447}{1000} x\right)$  (3)
> des := rhs(sol);
      des :=  $\frac{2000}{447} \sin\left(\frac{447}{1000} x\right) + \cos\left(\frac{447}{1000} x\right)$  (4)
> vel := diff(des, x);
      vel :=  $2 \cos\left(\frac{447}{1000} x\right) - \frac{447}{1000} \sin\left(\frac{447}{1000} x\right)$  (5)
> acel := diff(des, x$2);
      acel :=  $-\frac{447}{500} \sin\left(\frac{447}{1000} x\right) - \frac{199809}{1000000} \cos\left(\frac{447}{1000} x\right)$  (6)
> ur := 2.4504;
      ur := 2.4504 (7)
> vmax := 2.05;
      vmax := 2.05 (8)
> amax := 0.916;
      amax := 0.916 (9)
> plot([des, vel, acel, ur, vmax, amax], x = 0..20, y = -5..5, color = [blue, red, green, orange,
      orange, orange], legend = ["u(t)", "u'(t)", "u''(t)", "ur", "vmax", "amax"], title
      = "Vibración libre sin amortiguamiento", gridlines = true);

```

Vibración libre sin amortiguamiento

